SEMINAR REPORT

A Seminar Report on

AI IN DIGITAL FORENSICS

Submitted in partial fulfilment for the award of

THIRD YEAR DIPLOMA IN
COMPUTER ENGINEERING

By

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FOR THE ACADEMIC YEAR : 2026 – 2027

DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

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ABSTRACT

The rapid growth of computers, smartphones, cloud services, and the Internet has made digital technology an important part of our daily lives. At the same time, cybercrimes such as hacking, phishing, ransomware attacks, identity theft, online fraud, cyberbullying, and data breaches have also increased. Digital Forensics is the process of identifying, collecting, preserving, examining, analyzing, and presenting digital evidence from computers, mobile devices, networks, cloud systems, and other digital sources. It helps investigators solve cybercrime cases by finding and analyzing digital evidence. However, traditional digital forensic investigations mainly depend on manual analysis, which is time-consuming and becomes difficult when handling large amounts of digital data.

Artificial Intelligence (AI) is improving Digital Forensics by making investigations faster, smarter, and more accurate. AI technologies such as Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), and Computer Vision can automatically analyze digital evidence, detect suspicious activities, identify malware, classify files, recognize faces and objects in images and videos, and find hidden patterns in large datasets. These technologies reduce manual work, improve accuracy, and help investigators make better decisions. This seminar report explains the role of AI in Digital Forensics, its applications, advantages, limitations, and future scope. It also discusses how AI supports cybercrime investigations, malware detection, image and video analysis, network forensics, mobile forensics, and cloud forensics. The report concludes that AI is becoming an essential technology for modern digital forensic investigations and will continue to play an important role in strengthening cybersecurity and digital evidence analysis.

Keywords: Artificial Intelligence (AI), Digital Forensics, Cybercrime Investigation, Machine Learning, Digital Evidence, Cybersecurity.

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CHAPTER 1
INTRODUCTION

Digital Forensics is the process of identifying, collecting, preserving, examining, analyzing, and presenting digital evidence from computers, smartphones, networks, cloud systems, and other digital devices. As the use of digital technology has increased, cybercrimes such as hacking, phishing, ransomware attacks, identity theft, online fraud, cyberbullying, and data breaches have also increased. Digital Forensics helps investigators collect and analyze digital evidence to solve cybercrime cases and support legal investigations in courts of law.

Traditional digital forensic investigations mainly depend on manual analysis by forensic experts. Investigators examine hard disks, mobile phones, emails, network logs, memory dumps, databases, and system files to find useful evidence. This process is time-consuming and becomes difficult when handling large amounts of digital data generated from multiple devices and online platforms. It may also lead to delays, human errors, missed evidence, and increased investigation costs. As the volume and complexity of digital evidence continue to grow, traditional methods alone are no longer sufficient to handle modern cybercrime investigations efficiently.

Artificial Intelligence (AI) is transforming Digital Forensics by making investigations faster, smarter, and more accurate. AI technologies such as Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), and Computer Vision can automatically analyze digital evidence, detect malware, classify files, identify suspicious activities, recognize faces and objects in images and videos, recover deleted information, and discover hidden patterns in large datasets. AI can also prioritize important evidence, reduce false positives, and support investigators in making informed decisions. These technologies significantly reduce manual effort while improving the speed, accuracy, and reliability of forensic investigations.

AI is now being applied in many areas of Digital Forensics. In computer forensics, AI helps identify suspicious files and recover important evidence from storage devices. In mobile forensics, it analyzes call logs, messages, application data, images, and location history. In network forensics, AI monitors network traffic, detects intrusions, and identifies cyberattacks in real time. In cloud forensics, AI assists in collecting and analyzing evidence stored across cloud platforms. It is also widely used in malware analysis, email investigation, social media analysis, image and video forensics, and cyber threat intelligence.

For example, in a cybercrime case containing thousands of files, emails, chat messages, images, and videos, an AI-based forensic system can automatically filter irrelevant data, identify suspicious files, detect malicious software, recognize important keywords, and highlight critical evidence within a short time. Instead of manually examining every file, investigators can focus on the most relevant evidence, which speeds up investigations and improves overall efficiency.

Although AI provides many advantages, it also has certain challenges. The accuracy of AI systems depends on the quality of training data and algorithms. Issues such as privacy, legal acceptance of AI-generated evidence, explainability of AI decisions, and the possibility of biased results must also be considered. Therefore, AI should be used as a supporting tool for forensic experts rather than a complete replacement for human investigators.

The aim of this seminar is to study the role of Artificial Intelligence in Digital Forensics, understand its applications in cybercrime investigation, malware detection, image and video analysis, network forensics, mobile forensics, cloud forensics, and digital evidence management. It also aims to examine the advantages, limitations, and future scope of AI-based forensic techniques, and to explore how AI can improve the speed, accuracy, efficiency, and reliability of modern digital forensic investigations while strengthening cybersecurity and supporting the digital justice system.

CHAPTER 2
LITERATURE REVIEW

The research papers studied for this seminar are arranged according to different application areas of Artificial Intelligence (AI) in Digital Forensics. The selected papers focus on digital evidence mining, cybercrime investigation, malware detection, image forensics, and AI-assisted forensic analysis.

1. Digital Forensics AI: Evaluating, Standardizing and Optimizing Digital Evidence Mining Techniques (Solanke and Biasiotti, 2022)

This paper explains how Artificial Intelligence (AI) can improve digital forensic investigations by automatically identifying, extracting, and analyzing digital evidence. The authors discuss the use of Machine Learning (ML) for pattern recognition and evidence mining from large and complex digital datasets. The paper focuses on three important aspects of AI in Digital Forensics: evaluation, standardization, and optimization. It also proposes an AI-adaptive confidence scale to improve the reliability of AI-generated forensic results.

2. Machine Learning in Digital Forensics: A Systematic Literature Review (Nayerifard et al., 2023)

This paper presents a systematic review of research on Machine Learning applications in Digital Forensics published between 2010 and 2021. The authors analyze different forensic domains, including image forensics, malware analysis, network forensics, and mobile forensics. The review shows that image forensics has received the greatest benefit from Machine Learning techniques. It also highlights that Convolutional Neural Networks (CNNs) are the most widely used Deep Learning models in digital forensic research.

3. Digital Forensics and Strong AI: A Structured Literature Review (Fähndrich et al., 2023)

This paper studies the relationship between Artificial Intelligence and Digital Forensics by reviewing existing research on AI-based forensic methods. The authors explain that current digital forensic tools mainly use Weak AI, such as Machine Learning and expert systems, while Strong AI is not yet practically applied in forensic investigations. The paper discusses the

advantages of AI in evidence analysis and automation, while also highlighting the need for more fundamental research before advanced AI systems can be fully integrated.

4. A Comprehensive Analysis of the Role of Artificial Intelligence and Machine Learning in Modern Digital Forensics and Incident Response (Dunsin et al., 2024)

This paper presents a systematic review of the role of Artificial Intelligence and Machine Learning in modern Digital Forensics and Incident Response. It explains how AI can automate different stages of the forensic evidence lifecycle, including evidence identification, collection, preservation, examination, analysis, and reporting. The authors discuss the strengths and limitations of existing AI techniques and identify research gaps in automation, explainability, and forensic reliability. The paper concludes that AI and ML can significantly improve cybercrime investigations by reducing investigation time and increasing the accuracy of digital evidence analysis.

5. Artificial Intelligence in Mobile Forensics: A Survey of Current Status, a Use Case Analysis and AI Alignment Objectives (Scanlon et al., 2024)

This paper reviews the use of Artificial Intelligence (AI) in mobile forensic investigations. It explains how AI analyzes smartphone data such as messages, images, videos, call logs, and application records. AI techniques like Machine Learning and Computer Vision help investigators identify digital evidence quickly and accurately. The paper also discusses challenges such as transparency, reliability, privacy, and legal acceptance of AI-generated results. Overall, it concludes that AI can improve the speed and efficiency of mobile forensic investigations while requiring proper validation and human oversight.

6. Exploring the Potential of Large Language Models for Improving Digital Forensic Investigation Efficiency (Wickramasekara, Breitinger and Scanlon, 2025)

This paper explores how Large Language Models (LLMs) can improve the efficiency of digital forensic investigations. It explains that LLMs can assist investigators by analyzing digital evidence, summarizing forensic reports, answering investigation-related queries, and organizing large amounts of case data. These capabilities help reduce the time and effort required during investigations. The paper also discusses challenges such as hallucinations, bias, lack of explainability, privacy concerns, and legal acceptance. The authors conclude that LLMs

have strong potential to support digital forensics but should always be used with human expertise and proper validation to ensure accurate and reliable results.

7. Digital Forensics in Law Enforcement: A Case Study of LLM-Driven Evidence Analysis (2025)

This paper presents the use of Large Language Models (LLMs) in analyzing mobile chat data collected during real criminal investigations. The researchers evaluated GPT-4o, Gemini 1.5, and Claude 3.5 for understanding conversations, slang, abbreviations, and hidden meanings in digital evidence. The study found that LLMs can quickly summarize chats, identify relevant information, and reduce investigators' workload, leading to faster evidence analysis. However, the paper also emphasizes that AI-generated results must be verified by human forensic experts to ensure accuracy, reliability, and legal admissibility in court**.**

8. AI-Driven Digital Evidence Triage in Digital Forensics: A Comprehensive Review (Zamil and Khan, 2025)

This review paper explains how Artificial Intelligence (AI) helps investigators quickly identify, classify, and prioritize important digital evidence from large datasets. It discusses the use of Machine Learning (ML) and Deep Learning (DL) techniques for evidence classification, malware detection, image analysis, and automated data filtering. AI-based evidence triage reduces the time required to examine digital devices, allowing investigators to focus on the most relevant evidence first. The paper concludes that AI improves the speed, accuracy, and efficiency of digital forensic investigations while reducing manual effort, making it a valuable tool for modern cybercrime investigations.

9. ForensicLLM: A Local Large Language Model for Digital Forensics (Almadhoun et al., 2025)

This paper introduces ForensicLLM, a Large Language Model specially developed for digital forensic investigations. The model was trained using digital forensic research papers, forensic artifacts, and investigation-related knowledge to answer forensic questions more accurately. The study showed that ForensicLLM provides more relevant and reliable responses than general-purpose LLMs, helping investigators analyze digital evidence, generate reports, and support decision-making. Since it runs locally, it also improves data privacy and security, making it a useful tool for digital forensic experts.

10. LangurTrace: Forensic Analysis of Local LLM Applications (Kim et al., 2025)

This paper presents LangurTrace, a forensic framework designed to analyze digital artifacts generated by local Large Language Model (LLM) applications. The researchers identified important forensic evidence such as chat histories, uploaded files, generated documents, prompts, response logs, model configuration files, and application metadata. The framework provides a structured approach for collecting, preserving, and analyzing these artifacts during digital investigations. It also helps investigators understand user interactions with AI applications and recover valuable evidence from local systems. The authors conclude that LangurTrace improves the efficiency and reliability of forensic investigations involving LLM-based applications and provides a foundation for future AI forensic research.

11. Evaluating the Reliability of Digital Forensic Evidence Discovered by Large Language Models (IEEE COMPSAC, 2025)

This paper proposes a framework to improve the reliability of AI-generated digital evidence. It combines automated forensic artifact extraction with a Digital Forensic Knowledge Graph (DFKG) to validate evidence. The evaluation achieved over 95% accuracy in artifact extraction and demonstrated better traceability and consistency, making AI-assisted forensic investigations more reliable and suitable for legal use.

12. Digital Forensics in the Age of Large Language Models (Yin et al., 2025)

This paper explains how Large Language Models (LLMs) are transforming digital forensic investigations. It discusses the use of LLMs for evidence analysis, report generation, log analysis, and cybercrime investigations. The authors also describe challenges such as hallucinations, bias, explainability, and legal concerns. The paper concludes that LLMs have great potential in Digital Forensics when combined with human expertise and proper validation techniques.

Table No. 2.1: Literature Review

Sr. No.	Research Paper	Research Area	AI Technique	Major Contribution	Future Scope
1	Artificial Intelligence in Mobile Forensics (Scanlon et al., 2024)	Mobile Digital Forensics	AI, ML, NLP, Computer Vision	Improved analysis of smartphone evidence	More transparent and legally accepted AI models
2	Exploring the Potential of LLMs for Improving Digital Forensic Investigation Efficiency (2025)	Digital Forensic Investigation	Large Language Models (LLMs)	Faster evidence analysis and report generation	Better explainability and reduced hallucinations
3	Digital Forensics in Law Enforcement: A Case Study of LLM-Driven Evidence Analysis (2025)	Law Enforcement	GPT-4o, Gemini 1.5, Claude 3.5	Better understanding of criminal chat data	Improve reliability for courtroom use
4	AI-Driven Digital Evidence Triage in Digital Forensics (2025)	Evidence Triage	Machine Learning, Deep Learning	Faster identification and prioritization of evidence	Real-time triage for large-scale investigations
5	ForensicLLM: A Local Large Language Model for Digital Forensics (2025)	AI for Digital Forensics	Local LLM, NLP	Accurate forensic question answering with improved privacy	Expand training data and forensic knowledge
6	LangurTrace: Forensic Analysis of Local LLM Applications (2025)	AI Application Forensics	LLM Artifact Analysis	Identifies forensic artifacts from local LLM applications	Support more AI platforms and operating systems
7	AI-Based Malware Detection	Cyber Forensics	Machine Learning, Deep Learning	Improved malware detection accuracy	Detect emerging and zero-day threats

Sr. No.	Research Paper	Research Area	AI Technique	Major Contribution	Future Scope
8	AI for Image and Video Forensics	Multimedia Forensics	Computer Vision, CNN	Detects manipulated images and videos	Better detection of AI-generated deepfakes
9	Explainable AI (XAI) in Digital Forensics	Explainable AI	XAI Techniques	Makes AI decisions transparent and trustworthy	Standardized XAI methods for forensic tools
10	Future Trends of AI in Digital Forensics	AI-based Digital Forensics	AI, ML, LLMs	Highlights future developments in forensic automation	Stronger privacy, ethics, and legal compliance

CHAPTER 3
INFORMATION

3.1 Overview of AI in Digital Forensics

Artificial Intelligence (AI) in Digital Forensics refers to the use of AI techniques to assist in the collection, analysis, and interpretation of digital evidence. AI helps investigators examine large volumes of data from computers, smartphones, cloud storage, and IoT devices more quickly and accurately. Technologies such as Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Computer Vision, and Large Language Models (LLMs) automate tasks like evidence classification, malware detection, chat analysis, image and video examination, and forensic report generation.

3.2 Need for AI in Digital Forensics

Traditional digital forensic investigations require investigators to manually examine large amounts of digital evidence, making the process time-consuming and complex. The increasing use of smartphones, cloud services, social media, and IoT devices has created huge volumes of digital data that are difficult to analyze manually. The reviewed research papers show that AI helps overcome these challenges by automatically identifying important evidence, detecting suspicious activities, analyzing chat messages and multimedia files, and generating investigation reports. AI reduces investigators' workload, improves evidence analysis, and supports faster and more reliable cybercrime investigations.

3.3 Role of Artificial Intelligence in Digital Forensics

Artificial Intelligence (AI) plays a vital role in automating digital forensic investigations by analyzing large volumes of digital evidence quickly and accurately. The main AI technologies discussed in the reviewed papers are:

- Machine Learning (ML): Identifies patterns in digital evidence and supports malware detection, evidence classification, and anomaly detection.
- Deep Learning (DL): Analyzes complex data such as images, videos, and network traffic to detect manipulated content and cyber threats.
- Natural Language Processing (NLP): Examines text-based evidence such as emails, chat messages, and social media conversations to extract meaningful information.

3.4 Working Principle

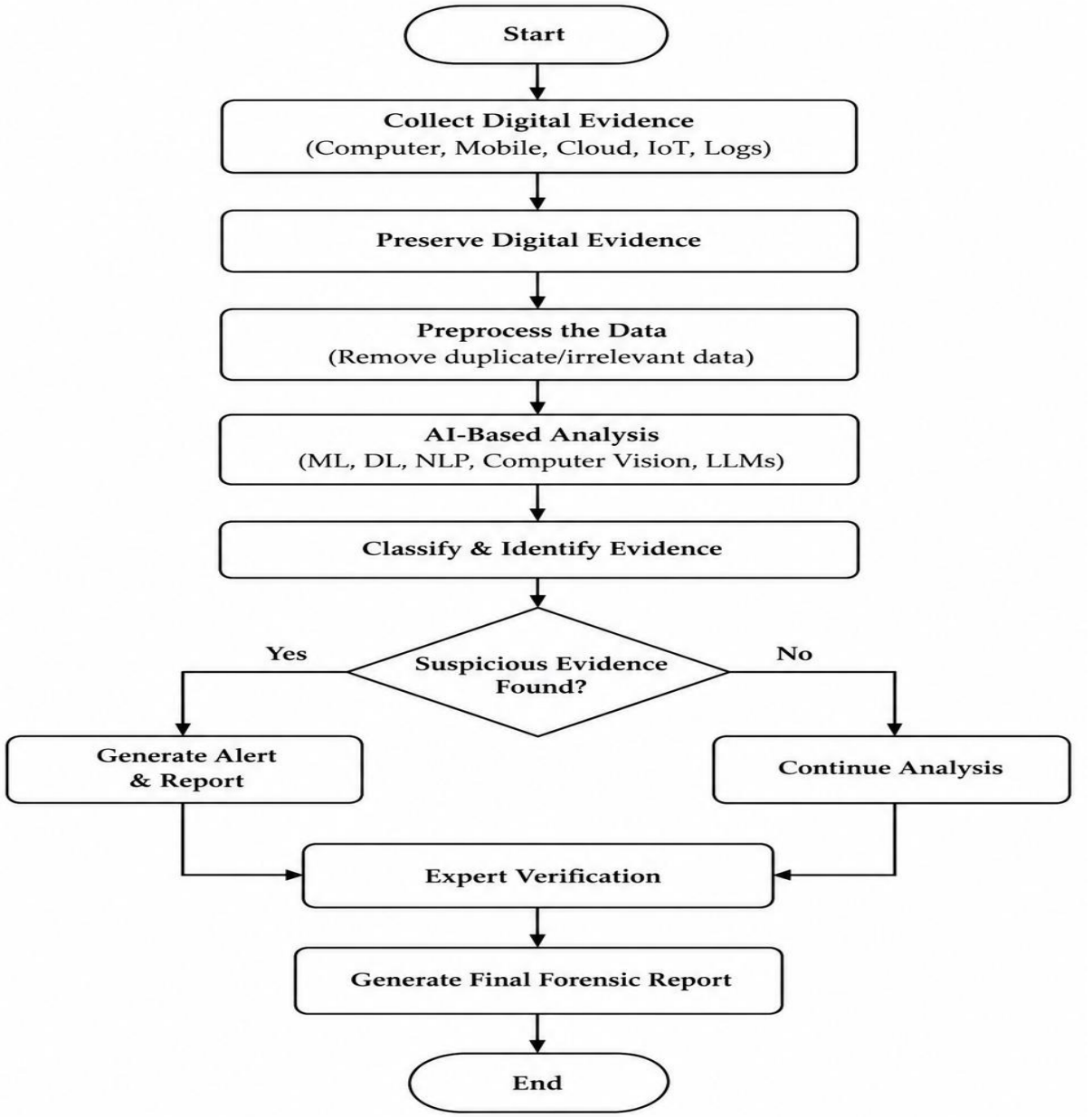


Fig. Working of AI in Digital Forensics System

The Digital Forensics System works in the following steps:

- **Collect Digital Evidence:** Evidence is collected from computers, smartphones, cloud storage, IoT devices, and network logs.
- **Preserve Evidence:** The collected evidence is securely preserved to maintain its integrity.
- **Preprocess Data:** Duplicate and irrelevant data are removed, and the evidence is prepared for analysis.

- **AI-Based Analysis:** AI techniques such as Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), and Large Language Models (LLMs) analyze the digital evidence.
- **Evidence Identification:** The system identifies and classifies relevant or suspicious evidence.
- **Report Generation:** AI generates summaries and investigation reports from the analyzed evidence.
- **Expert Verification:** Forensic experts verify the AI-generated findings for accuracy and legal admissibility.
- **Final Report:** The verified evidence is compiled into the final forensic investigation report for further investigation or court proceedings.

3.5 Applications of AI in Digital Forensics

The reviewed research papers show that AI is applied in many areas of digital forensic investigations, including:

- Mobile Forensics: Examining data from smartphones, including messages, call logs, photos, videos, and applications.
- Malware Analysis: Detecting and classifying malicious software using AI-based techniques.
- Image and Video Forensics: Identifying manipulated images, deepfakes, and edited videos.
- Digital Evidence Triage: Automatically filtering and prioritizing important evidence from large datasets.
- Forensic Report Generation: Using LLMs to summarize investigation findings and prepare forensic reports.
- Cybercrime Investigation: Supporting law enforcement agencies in analyzing digital evidence and identifying cybercriminal activities.

3.6 Classification of Digital Evidence

Digital evidence is collected from different digital devices and online platforms during forensic investigations. Artificial Intelligence classifies this evidence into different categories, making

it easier for investigators to analyze relevant information and identify cybercrime activities quickly and accurately.

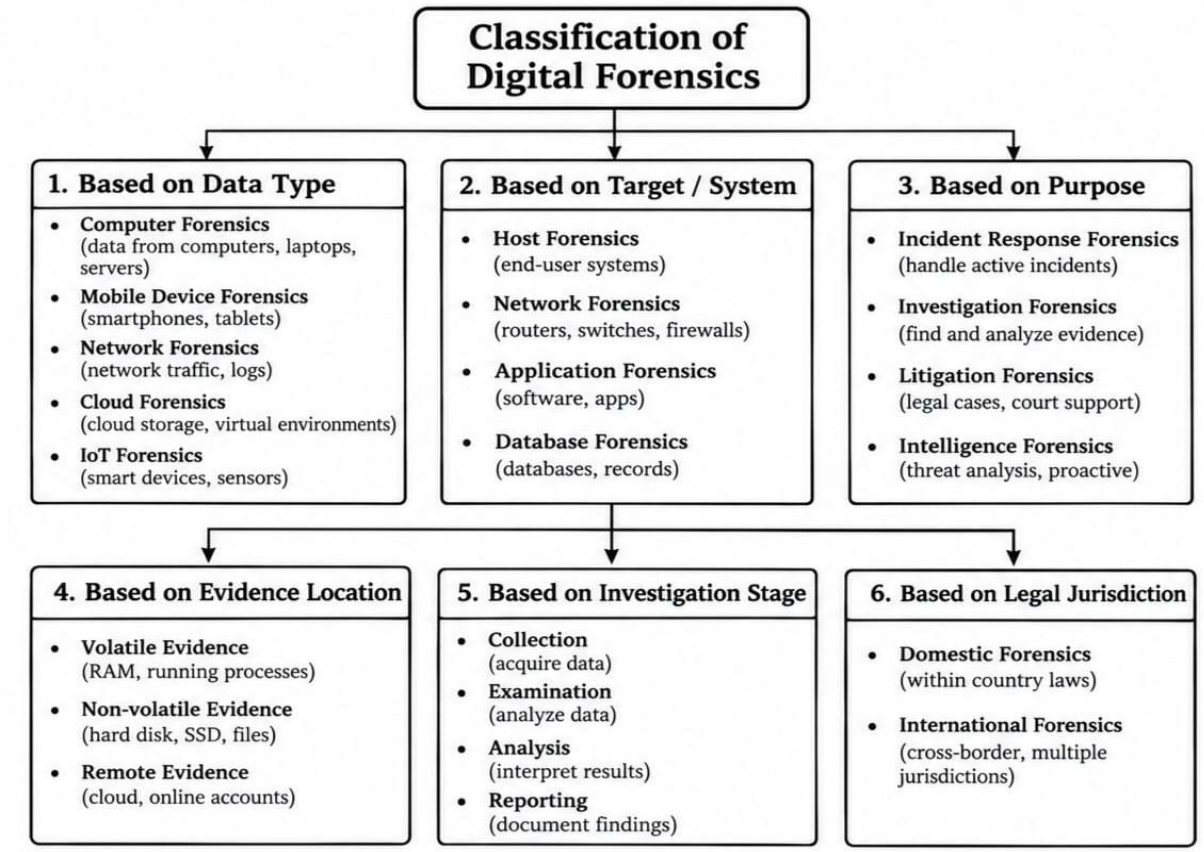


Fig.3.2 Classification of Digital Forensics

- **Description:**
  - **Documents & Files:** PDF, Word, Excel, text files, and system documents.
  - **Images & Videos:** Photos, screenshots, CCTV footage, and multimedia evidence.
  - **Chats & Emails:** WhatsApp, Telegram, SMS, emails, and social media conversations.
  - **Network Logs:** IP addresses, firewall logs, server logs, and network traffic.
  - **Cloud Storage:** Google Drive, OneDrive, Dropbox, cloud backups, and online documents.

CHAPTER 4
ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. AI significantly reduces the time required to analyze large volumes of digital evidence.
2. Machine Learning automatically classifies and prioritizes important digital evidence.
3. Large Language Models (LLMs) assist investigators in summarizing reports and analyzing documents.
4. AI improves the accuracy of malware detection, image analysis, and cybercrime investigations.
5. AI helps identify hidden patterns and suspicious activities that may be difficult to detect manually.
6. Automation reduces investigators' workload and improves overall forensic efficiency.
7. AI supports faster decision-making and enhances the quality of digital forensic investigations.

4.2 Disadvantages:

1. AI models may generate incorrect or misleading results (hallucinations), especially in LLMs.
2. The accuracy of AI depends on the quality and availability of training data.
3. AI decisions may lack transparency and explainability in some cases.
4. Human forensic experts are still required to verify AI-generated findings.
5. Privacy and security concerns arise when sensitive forensic data is processed.
6. Legal and ethical issues may affect the acceptance of AI-generated evidence in court.
7. Developing and maintaining AI-based forensic systems requires specialized technical knowledge and computational resources.

CHAPTER 5
CONCLUSION

The study of the selected research papers on AI in Digital Forensics shows that Artificial Intelligence has significantly improved the speed, accuracy, and efficiency of digital forensic investigations. AI technologies such as Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Computer Vision, and Large Language Models (LLMs) help investigators analyze large volumes of digital evidence, detect cyber threats, classify forensic artifacts, identify suspicious activities, and generate forensic reports more efficiently than traditional methods.

The reviewed papers demonstrate that AI is widely applied in mobile forensics, malware analysis, digital evidence triage, image and video forensics, forensic report generation, and law enforcement investigations. These technologies reduce manual effort, improve evidence analysis, and assist investigators in making faster and more informed decisions. AI-based frameworks such as ForensicLLM and LangurTrace further highlight the growing role of domain-specific AI models in supporting forensic investigations while maintaining data privacy and improving evidence management.

Overall, the literature review indicates that AI has the potential to transform digital forensic investigations by automating evidence analysis, improving investigation efficiency, and supporting cybercrime detection. With further research, better AI models, explainable AI techniques, and standardized forensic frameworks, AI is expected to become an essential component of future digital forensic systems and law enforcement investigations.

CHAPTER 6
PROJECT INITIATION

6.1 Description of Problem Statement

The rapid growth of computers, smartphones, cloud services, social media, and IoT devices has resulted in a massive increase in digital data. During cybercrime investigations, forensic experts must examine large volumes of digital evidence such as emails, chat messages, images, videos, system logs, and application data. Traditional digital forensic investigations rely heavily on manual analysis, making the process time-consuming, complex, and prone to human error.

6.2 Scope and Objectives:

Scope of the Project: The project aims to study the application of Artificial Intelligence (AI) in Digital Forensics through recent research papers and analyze how AI techniques improve digital forensic investigations. It focuses on AI technologies such as Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Computer Vision, and Large Language Models (LLMs) for evidence analysis, malware detection, digital evidence triage, multimedia analysis, and forensic report generation. The study also examines the benefits, challenges, limitations.

Objectives of the Project

To study the role of Artificial Intelligence in digital forensic investigations through recent research papers. To understand the application of Machine Learning, Deep Learning, NLP, Computer Vision, and LLMs in digital forensics. To analyze how AI improves digital evidence collection, classification, and analysis. To study AI-based techniques for malware detection, mobile forensics, and multimedia forensics. To examine the use of Large Language Models (LLMs) in forensic report generation and evidence analysis.

- **6.3 Stakeholders**

The stakeholders of the **AI in Digital Forensics** project are the individuals and organizations involved in developing, using, managing, or benefiting from AI-assisted digital forensic investigations.

Table No. 6.1: Stakeholders

Sr. No.	Stakeholder	Role / Interest
1	Digital Forensic Investigators	Collect, analyze, and interpret digital evidence using AI tools.
2	Law Enforcement Agencies	Use AI-assisted forensic investigations to solve cybercrime cases.
3	Cybersecurity Professionals	Detect cyber threats, malware, and security incidents using AI techniques.
4	Courts and Legal Authorities	Use verified digital evidence during legal proceedings and investigations.
5	AI Developers / Researchers	Develop and improve AI models and forensic analysis tools.
6	Organizations / Companies	Protect digital assets and investigate cyber attacks and data breaches.
7	Project Guide and Evaluators	Guide, monitor, and evaluate the seminar/project work.

6.4 Platform / Equipment / Resources Identification:

1. Platform:

The study of AI in Digital Forensics is based on recent research papers, AI technologies, and digital forensic tools. The project can be carried out on Windows or Linux operating systems using Python for AI-based analysis. Development and experimentation can be performed using Visual Studio Code or Jupyter Notebook. AI models such as Machine Learning (ML), Deep Learning (DL), and Large Language Models can be implemented using Python.

2. Equipment:

Table No. 6.2: Equipment

Sr. No.	Equipment	Purpose
1	Computer / Laptop	Run AI models and digital forensic tools.

2	Smartphone	Source of mobile forensic evidence.
3	External Storage Device	Store and transfer digital evidence securely.
4	Internet Connection	Access research papers and AI resources.
5	Digital Forensic Workstation	Analyze digital evidence efficiently (if available).
6	Cloud Storage (Optional)	Store forensic datasets and reports.
7	USB Drive	Collect and transfer digital evidence.

3. Resources:

Table No. 6.3: Resources

Resources	**Name**	**Purpose**
Operating System	Windows 11 / Linux	Run forensic software and AI applications.
Programming Language	Python	AI model development and data analysis.
AI Libraries	TensorFlow, PyTorch, Scikit-learn, Pandas	Machine Learning and Deep Learning.
Large Language Models	GPT-4o, Gemini, Claude, ForensicLLM	Evidence analysis and report generation.
Digital Forensic Tools	Autopsy, FTK Imager, Cellebrite (if available)	Collection and analysis of digital evidence.
Development Environment	Jupyter Notebook / Visual Studio Code	Coding, testing, and experimentation.
Database	MySQL / SQLite	Store forensic information and analysis results.
Documentation Tools	Microsoft Word, PowerPoint	Prepare reports and seminar presentation.
Research Resources	IEEE Xplore, Springer, ScienceDirect, Google Scholar	Study recent research papers on AI in Digital Forensics.

CHAPTER 7
SEMINAR PRESENTATION

	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2026-2027*

SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	**SPI – 315003**	**Date : 17/10/2026**
Topic : AI in Digital Forensics		

Roll No.	Name of Student
75	Vedika Hemant Patil
87	Trupti Chhotu Shende
94	Kirtika Suresh Sonwane
109	Neha Gangadhar Wagh

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Abstract

- AI in Digital Forensics uses Artificial Intelligence (AI) and Machine Learning (ML) to assist investigators in examining large volumes of digital evidence from computers, mobile phones, networks and cloud services. Traditional investigations can require significant time to search files, messages, images and logs.
- AI can help classify files, detect suspicious patterns, group related evidence and reconstruct event timelines. Research shows promising support for evidence triage and analysis, but AI results must be validated by trained investigators.
- The proposed seminar project explores a prototype that analyzes a prepared forensic dataset, flags potentially relevant artifacts and produces a reviewable report while preserving evidence integrity and human oversight.

3

1. Introduction

- Digital forensics is the process of identifying, collecting, preserving, examining and presenting digital evidence.
- Digital evidence may come from computers, mobile phones, networks, cloud accounts and multimedia files.
- Manual examination of large datasets can be slow and difficult.
- AI and ML can help prioritize files, recognize patterns and identify suspicious activity.



4

2. Literature Survey

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	AI/ML in digital forensics review	Machine learning, pattern recognition, forensic workflow	Literature review	Survey AI use across evidence collection, analysis and incident response	Research maturity varies by task; validation needed.
2	AI in mobile forensics survey	AI/ML, mobile-device artifacts	Literature review	Review current AI uses and mobile forensic scenarios	Models may not generalize to real-world data.
3	AI-assisted evidence triage	Classification, ranking, keyword/rule matching	Prepared forensic files	Prioritize artifacts for human examination	False positives and missed evidence are possible.
4	Timeline reconstruction support	Log parsing, event correlation, ML	System and application logs	Connect events and help build investigation timelines	Incomplete timestamps and noisy logs affect results.
5	Multimedia forensic analysis	Computer vision, image/audio classification	Image, audio and video samples	Group media and flag suspicious or manipulated content	Performance depends on data quality and model bias.
6	Malware and file classification	Static features, ML classifiers, file hashes	Known benign/malicious samples	Identify files needing closer examination	New or obfuscated malware can evade detection.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	Network incident forensics	Log analysis, anomaly detection, ML	Network-flow and security logs	Find unusual traffic and correlate events	Encrypted traffic and false alerts limit analysis.
8	Evidence integrity and handling	SHA-256 hashing, audit logs, chain of custody	Forensic images and evidence records	Check integrity and document evidence handling	AI does not replace established evidence procedures.
9	Automated artifact extraction	Forensic parsers, Python, databases	Disk/mobile forensic artifacts	Extract structured records for review	Tool support varies by device and file system.

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Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	Forensic tool testing	Repeatable test sets, validation procedures	Known/reference datasets	Assess tool behavior and reproducibility	Testing one scenario does not prove universal reliability.
11	AI-supported reporting	Summarization, structured reports, analyst review	Verified investigation findings	Make results easier to review and communicate	Generated text must be checked against source evidence.

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3. Information to seminar topic

- **Technology:** Artificial Intelligence (AI), Machine Learning (ML) and Digital Forensics
- **Main Purpose:** To assist investigators in finding relevant digital evidence efficiently.
- **Key Concept:** AI analyzes patterns in files, logs, messages and multimedia data.
- **Approach:** Collect forensic images/logs safely, extract artifacts, analyze copies using rules and ML, then verify findings.
- **Main Benefit:** Faster evidence triage and better prioritization of suspicious artifacts.
- **Application:** Cybercrime investigations, incident response, mobile/device forensics and fraud analysis.

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4. Advantages / Disadvantages

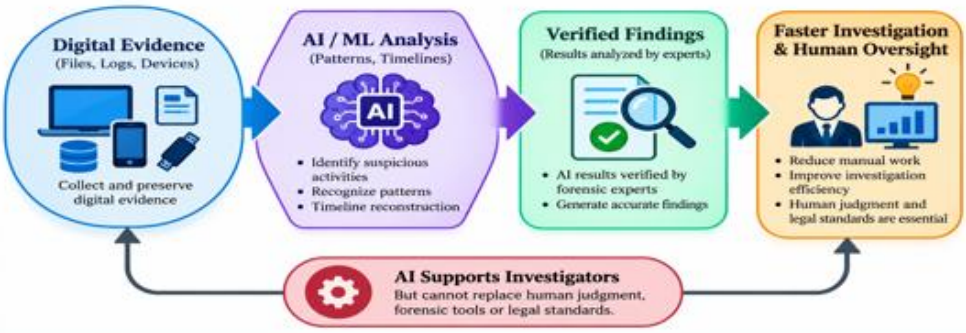
- **Advantages**
- Speeds up searching and sorting large volumes of digital evidence.
- Helps identify unusual patterns, suspicious files and related artifacts.
- Can support timeline reconstruction and evidence prioritization.
- Reduces repetitive manual work for investigators.
- Can assist analysis of text, images, logs and network events.

- **Disadvantages**
- AI may produce false positives or miss relevant evidence.
- Results can be biased if training data is incomplete.
- Some models are difficult to explain or reproduce.
- Sensitive evidence requires strong privacy and access controls.
- AI findings alone may not be sufficient for legal conclusions.

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5. Conclusion

AI helps digital forensic investigators analyze large amounts of digital evidence quickly and identify suspicious activities. It also supports pattern recognition, timeline reconstruction and incident response. However, evidence must be preserved properly, and AI results must be verified by experts. AI can reduce manual work and improve investigation efficiency. Therefore, AI supports investigators but cannot replace human judgment, forensic tools or legal standards.



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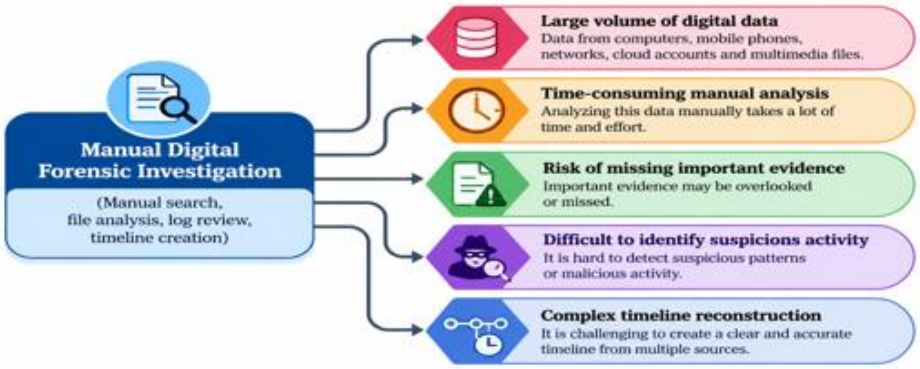
6. Project Initiation

- The AI in Digital Forensics seminar was initiated to explore how intelligent techniques can support investigations involving computers, mobile devices and other digital sources.
- The proposed project focuses on analyzing a controlled sample dataset, extracting useful artifacts, identifying suspicious patterns with AI/ML assistance and presenting findings in a simple report.
- The design emphasizes evidence integrity, transparent results and human verification.

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6.1 Description of Problem Statement

Digital investigations generate large amounts of data from computers, mobile phones, networks, cloud accounts and multimedia files. Analyzing this data manually is time-consuming, complex and prone to human error. Important evidence may be missed, and it is difficult to identify suspicious activity or create a clear timeline. Therefore, there is a need for an AI-based system to automatically analyze and prioritize digital evidence to support investigators.



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6. 2 Scope and objectives

Objectives

- Understand the digital forensic evidence lifecycle.
- Load a prepared sample dataset or forensic artifacts for analysis.
- Use Python-based rules and ML/NLP techniques to categorize or prioritize artifacts.
- Flag potentially suspicious patterns for investigator review.
- Generate a report containing findings, timestamps and confidence/limitations.

Scope

- Educational prototype using lawful, synthetic or publicly available sample data.
- Focus on evidence triage and analysis—not autonomous accusations.
- Possible tools: Python, pandas, scikit-learn, SQLite, Autopsy and hashing utilities.
- Results require human verification; the prototype is not a certified forensic tool.

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6. 3 Stake holder

- **Digital Forensic Investigator:** Defines the investigation and validates findings.
- **Cybersecurity / Incident Response Team:** Uses findings to understand suspicious activity.
- **Law Enforcement and Legal Professionals:** May review evidence and reports under applicable procedures.
- **Organization / Evidence Owner:** Authorizes access and protects sensitive data.
- **Project Developers and Guide:** Build, test and document the prototype.
- **Researchers and Students:** Study AI methods, limitations and responsible evidence handling.

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6. 4 Seminar exsicussion plan

1. **Current Scenario:** Digital evidence is collected from computers, mobiles, networks and cloud services.
1. **Investigation Challenges:** Large data, deleted files, different formats and limited time.
1. **Existing Approaches:** Forensic tools, keyword searches, file hashes and manual analysis.
1. **Research Gap:** AI can help analyze evidence, but accuracy and reliability are important.
1. **Proposed System:** A Python-based system to identify important evidence and generate reports.
1. **Evaluation:** Test the system using sample cases and check errors.
1. **Future Work:** Add more evidence types and improve AI accuracy.

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6.5 Platform/ Equipment/ Resources

- **Hardware:** Computers, laptops, mobile devices and storage devices are required to collect and examine digital evidence.
- **Software Tools:** Digital forensic tools such as Autopsy, FTK and EnCase are used to analyze and recover digital evidence.
- **AI / ML Platforms:** Python, TensorFlow and Scikit-learn help in pattern recognition, data classification and suspicious activity detection.
- **Data Storage:** Hard disks, SSDs and secure cloud storage are used to preserve digital evidence safely.
- **Operating System:** Windows and Linux provide platforms for running forensic tools and analyzing digital data.
- **Network Resources:** Network monitoring tools help investigate network traffic and identify suspicious activities.
- **Security Resources:** Encryption, access control and hashing techniques help maintain evidence integrity and security.
- **Human Resources:** Digital forensic investigators and AI experts verify results and ensure accurate investigation.

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Thank you

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SEMINAR AND PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Seat Number	Name of Student
1	24610960258	388562	Patil Vedika Hemant
2	24610960273	388574	Shende Trupti Chhotu
3	24610960281	388581	Sonawane Kirtika Suresh
4	24610960297	388596	Wagh Neha Gangadhar

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388562	388574	388581	388596
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : Mr. Nishad D.Patel

(Course Expert/s)